Context Dynamic Injection

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1. Overview: Layered Architecture of Context Injection

Claude Code's dynamic context injection is not a single mechanism, but consists of **13 independent subsystems**, each with clear responsibilities and well-defined boundaries, ultimately converging into API calls through a unified message construction pipeline. The overall architecture can be summarized as:

```
API call (query())
  ↑
  ┌─ systemPrompt (from prompts.ts / systemPromptSections.ts)
  ├─ userContext (from context.ts: getUserContext)
  ├─ systemContext (from context.ts: getSystemContext)
```

```
  ├─ messages[] (user messages + system attachment messages)
  └─ attachments (skills, agents, mcp, tool delta, etc.)
```

---

2. Layer 1: System Reminder Dynamic Injection

2.1 Core Mechanism

System Reminder is Claude Code's most fundamental context injection mechanism. By wrapping system-level information in `<system-reminder>...</system-reminder>` XML tags, it **injects system-level information inside user messages**, bringing it into the model's attention context.

Core function: `src/utils/messages.ts`

```
// Entry function
export function wrapInSystemReminder(content: string): string {
  return `<system-reminder>\n${content}\n</system-reminder>`
}

// Batch wrapping
export function wrapMessagesInSystemReminder(
  messages: Message[],
): Message[]
```

2.2 Injection Scenarios

All `system/init` type messages built via `buildSystemInitMessage()`, as well as attachment messages generated by various attachment building functions, are automatically wrapped in `<system-reminder>` tags:

```
// src/utils/messages.ts attachment wrapping example
return wrapMessagesInSystemReminder([
  createAttachmentMessage({
    type: 'system_local_command',
    content: `${formatShellPrefixCommand(command, workingDirectory)} ${input}`,
  }),
])
```

2.3 Injection Content Classification

The system prompt text itself (`prompts.ts`) also contains instructions about System Reminders:

```
function getSystemRemindersSection(): string {
  return `- Tool results and user messages may include <system-reminder> tags.
    <system-reminder> tags contain useful information and reminders. They are
    automatically added by the system, and bear no direct relation to the specific
    tool results or user messages in which they appear.
  - The conversation has unlimited context through automatic summarization.`
}
```

2.4 System Reminder Collection in Beta Tracing

In `src/utils/telemetry/betaSessionTracing.ts`, the system extracts all system-reminder content from messages for telemetry:

```
function extractSystemReminderContent(text: string): string | null {
  const match = text.match(/<system-reminder>([\s\S]*?)<\/system-reminder>/)
  return match ? match[1] : null
}
```

2.5 Message Types Wrapped in `<system-reminder>`

All messages processed through `normalizeAttachmentForAPI()`, after being wrapped by `wrapMessagesInSystemReminder()`, carry `<system-reminder>` tags in two forms:

- String content: wrapped as a whole as `<system-reminder>\n{content}\n</system-reminder>`
- Array content: only blocks with `type: 'text'` are wrapped; other block types like `type: 'tool_result'` remain unchanged

Content types that are actually wrapped (attachments):

类型	内容说明
plan_mode / plan_mode_reentry / plan_mode_exit	计划模式工作流指令
auto_mode / auto_mode_exit	自动模式指令
edited_text_file	文件变更通知
compact_file_reference	摘要后旧文件引用
pdf_reference	大 PDF 轻量引用
selected_lines_in_ide	IDE 选中行
opened_file_in_ide	IDE 打开文件通知
plan_file_reference	计划文件内容
invoked_skills	已调用技能内容
todo_reminder	TodoWrite 提醒
task_reminder	TaskCreate/TaskUpdate 提醒
nested_memory / relevant_memories	记忆内容
skill_listing / skill_discovery	技能列表/发现
queued_command	本地排队命令
output_style	输出风格通知
diagnostics	诊断问题
mcp_resource	MCP 资源内容
agent_mention	Agent 提及
task_status	后台任务状态
async_hook_response	Hook 异步响应
token_usage / budget_usd / output_token_usage	限额统计
hook_blocking_error / hook_success / hook_additional_context / hook_stopped_continuation	Hook 事件
compaction_reminder / context_efficiency	自动压缩提示
date_change	日期变更
ultrathink_effort	推理 effort 级别
deferred_tools_delta / agent_listing_delta / mcp_instructions_delta	增量更新通知
companion_intro	Companion 介绍
verify_plan_reminder	VerifyPlanExecution 调用提醒
critical_system_reminder	任意关键提醒
teammate_mailbox / team_context	Swarm 多智能体消息

Design Philosophy: All these entries are information that "the model needs to act upon, but should not directly tell the user about."

Categories:

Direct instructions that change model behavior

`plan_mode`, `auto_mode`, `verify_plan_reminder`, `ultrathink_effort` — the user has switched to a certain working mode, and the system needs to notify the model "how you should do things now"

User-known system state changes

`edited_text_file`, `selected_lines_in_ide`, `opened_file_in_ide`, `date_change` — the user edited a file themselves, selected lines themselves; the model needs to know these happened, but shouldn't say "I noticed you edited the file" in its response

Background knowledge that tools cannot proactively retrieve

`invoked_skills`, `relevant_memories`, `nested_memory`, `mcp_resource` — the user didn't request this information via a tool; it was injected by the system proactively

System-internal state / warnings

`diagnostics`, `compaction_reminder`, `context_efficiency`, `token_usage`, `budget_usd` — fully internalized system signals; even if the model knows about them, it should not tell the user, nor should it explain them to the user

Hook event streams

`hook_blocking_error`, `hook_success`, `hook_additional_context`, etc. — feedback on hook execution results, for the model's decision-making

---

3. Layer 2: CLAUDE.md / Memory File Injection

3.1 File Discovery Hierarchy

`src/utils/claudemd.ts` implements a complete CLAUDE.md file discovery and injection pipeline. Files are loaded in the following priority order (lower priority loaded first, higher priority loaded later; the model pays more attention to later-loaded content):

```
Loading order (low to high priority):
1. Managed (global management level, /etc/claude-code/CLAUDE.md)
2. User (user global level, ~/.claude/CLAUDE.md)
3. Project (project level, CLAUDE.md / .claude/CLAUDE.md / .claude/rules/*.md found by
traversing upward)
4. Local (local level, CLAUDE.local.md)

Finally merged into a single string, concatenated segment by segment in the form of
"Contents of <path>:"
```

3.2 Key Functions

```
// Main entry: discover and load all memory files
export const getMemoryFiles = memoize(
  async (forceIncludeExternal: boolean = false): Promise<MemoryFileInfo[]>
)

// Convert file list to system prompt string
export const getClaudeMds = (memoryFiles: MemoryFileInfo[]): string => {
  // Adds descriptive suffixes for each type, formatted as:
  // "Contents of /path/to/CLAUDE.md (project instructions, checked into the
```

```
codebase):\n\n<content>"
}
```

3.3 @ Directive Injection

Memory files support including other files via the `@path` syntax:

```
// @include supported path formats
// @path, @./relative/path, @~/home/path, @/absolute/path
// Only expanded in plain text nodes; not expanded inside code blocks or HTML comments
// Maximum recursion depth: 5 levels
```

> **Note**
>
> `@path` references in memory files are expanded at load time, injecting the content of the target file

3.4 HTML Comment Stripping

When parsing memory file content, block-level HTML comments (`<!-- ... -->`) are automatically stripped:

```
// src/utils/claudemd.ts
export function stripHtmlComments(content: string): { content: string; stripped: boolean }
```

> **Note**
>
> - HTML comments manually written by the author in CLAUDE.md are essentially human-only annotations
> - **This corresponds to content the author wrote in CLAUDE.md but doesn't want the AI to see — annotations visible to humans but hidden from the AI**

Typical Scenarios:

```
1. Internal TODO / checklist annotations
<!-- TODO: This rule needs to be rewritten after v2 release -->
<!-- FIXME: Temporary workaround, remove this rule in production -->

2. Draft content (not yet finalized)
<!-- The spec below is still under discussion and not yet in effect -->
## Experimental Feature Handling
Still confirming with the team...

3. Instructions only needed by certain people
<!-- Note for maintainers: This directory structure will be refactored next version -->
## Deployment Configuration

4. Context comments in multi-person collaboration
<!-- 2024-06 Added this rule after security review -->
Never concatenate user input directly into SQL.

<!-- legal-team-2024-07: This exception clause was added per legal requirements -->

5. Conditional comments (debug / development only)
```

```
  <!-- DEBUG: Skip certificate verification during local development -->
  When connecting to external services, verify certificates...
```

Design Intent

The author wants to preserve these human collaboration traces in CLAUDE.md (review records, TODOs, draft content), but does not want them to enter the AI context — because:

- TODOs/drafts can confuse the AI
- Internal analysis comments pollute the context
- Temporary workarounds should not influence the AI's judgment

The stripping logic is designed as: **if a comment is written in block-level form (on its own line), it is assumed to be a human annotation by default**; if written inline within text (using the comment syntax as part of the prose), it is meaningful text that should not be stripped

3.5 MEMORY.md Entrypoint Truncation

The `MEMORY.md` entry file has dual limits:

```
// src/memdir/memdir.ts
export const MAX_ENTRYPOINT_LINES = 200   // Maximum 200 lines
export const MAX_ENTRYPOINT_BYTES = 25_000 // Maximum 25KB

// A warning is appended when truncating:
// "> WARNING: MEMORY.md is {reason}. Only part of it was loaded..."
```

> **Note**
>
> MEMORY.md is the memory index; it only stores pointers, in the format `- [Title](filename.md) — brief description`

3.6 Conditional Rules

Files can restrict their applicable scope via the frontmatter `paths` field:

```
---
paths:
  - src/**/*.ts
  - src/**/*.tsx
---
Your instructions here...
```

```
// src/utils/claudemd.ts
export async function processConditionedMdRules(
  targetPath: string,
  rulesDir: string,
  type: MemoryType,
  processedPaths: Set<string>,
  includeExternal: boolean,
): Promise<MemoryFileInfo[]>
// Uses the ignore library for glob pattern matching
```

The system matches the current working directory path via `glob` patterns, injecting the memory only when matched.

> **Note**
>
> A typical **context-aware** or **dynamic prompt injection** design. Simply put:
>
> - Normal memories are "globally active" — once loaded, they always exist in context
> - Conditional rules allow a memory file to declare: **"this memory is only useful when I'm editing files matching specific paths, so inject it only when matched"**

3.7 Integration with `context.ts`

`getUserContext()` in `src/context.ts` calls `getClaudeMds()`:

```
// src/context.ts
export const getUserContext = memoize(async (): Promise<{ [k: string]: string }> => {
  const claudeMd = getClaudeMds(filterInjectedMemoryFiles(await getMemoryFiles()))
  setCachedClaudeMdContent(claudeMd || null) // Cache for use by other modules
  return {
    ...(claudeMd && { claudeMd }),
    currentDate: `Today's date is ${getLocalISODate()}.`,
  }
})
```

3.8 Cache Invalidation Mechanism

```
// Reset cache (trigger full reload)
export function resetGetMemoryFilesCache(reason: InstructionsLoadReason): void

// Clear cache only (no hook trigger; for correctness purposes only)
export function clearMemoryFileCaches(): void

// loadReason propagation: 'session_start' | 'compact' | 'include'
let nextEagerLoadReason: InstructionsLoadReason = 'session_start'
```

> **Note**
>
> These two functions serve **hook notification + cache invalidation coordination**. Their relationship to dynamic injection is: **letting the hook caller know why memory was reloaded**, so it can decide how to respond (e.g., after compaction the context needs to be rebuilt, while after a worktree switch the cache just needs to be silently refreshed)

---

4. Layer 3: Auto Memory System

> **Note**
>
> **See "Memory System" documentation for details**

4.1 Overview

Auto Memory is a complete file-based persistent memory system that injects guiding prompts into the System Prompt via `loadMemoryPrompt()`. Core file: `src/memdir/memdir.ts`.

4.2 Memory Type Classification

The system mandates 4 memory types (`src/memdir/memoryTypes.ts`):

```
export const MEMORY_TYPES = ['user', 'feedback', 'project', 'reference'] as const
```

Each type has explicit `when_to_save`, `how_to_use`, and `body_structure` specifications.

4.3 Persistence Structure

```
~/.claude/projects/<slug>/memory/
  MEMORY.md          # Entrypoint index (one link per entry)
  user_role.md       # Topic file (frontmatter + content)
  feedback_*.md
  project_*.md
  reference_*.md
```

4.4 Memory File Frontmatter Format

```
---
name: {{memory name}}
description: {{one-line description}}
type: {{user, feedback, project, reference}}
---
{{memory content}}
```

4.5 Integration with System Prompt

```
// src/memdir/memdir.ts
export async function loadMemoryPrompt(): Promise<string | null> {
  // 1. KAIROS daily log mode (append-only)
  if (feature('KAIROS') && autoEnabled && getKairosActive()) {
    return buildAssistantDailyLogPrompt(skipIndex)
  }
  // 2. TEAMMEM team memory mode
  if (feature('TEAMMEM') && teamMemPaths!.isTeamMemoryEnabled()) {
    return teamMemPrompts!.buildCombinedMemoryPrompt(extraGuidelines, skipIndex)
  }
  // 3. Standard auto memory
  if (autoEnabled) {
    return buildMemoryLines('auto memory', autoDir, extraGuidelines, skipIndex).join('\n')
  }
  return null
}
```

4.6 Behavioral Constraint Guidance

The System Prompt injected by Auto Memory includes a detailed "what NOT to save" list:

```
// src/memdir/memoryTypes.ts
export const WHAT_NOT_TO_SAVE_SECTION: readonly string[] = [
  '## What NOT to save in memory',
  '- Code patterns, conventions, architecture, file paths, or project structure...',
  '- Git history, recent changes, or who-changed-what...',
  '- Debugging solutions or fix recipes...',
  '- Anything already documented in CLAUDE.md files.',
  '- Ephemeral task details...',
  // Important: filter even when the user explicitly asks to save
  'These exclusions apply even when the user explicitly asks you to save.'
]
```

4.7 Stale Memory Verification Guidance

```
// src/memdir/memoryTypes.ts
export const TRUSTING_RECALL_SECTION: readonly string[] = [
  '## Before recommending from memory',
  'A memory that names a specific function, file, or flag is a claim that it existed...',
  '- If the memory names a file path: check the file exists.',
  '- If the memory names a function or flag: grep for it.',
  '"The memory says X exists" is not the same as "X exists now."'
]
```

---

5. Layer 4: Session Memory

> **Note**
>
> **See "Context Management" documentation for details**

5.1 Overview

Session Memory is a **session-level** memory mechanism that automatically maintains a Markdown file during a conversation, recording key information from the current session. Core file: `src/services/SessionMemory/sessionMemory.ts`.

5.2 Trigger Mechanism

Session Memory is triggered via **Post-Sampling Hook** (see Layer 9), running automatically after model sampling completes:

```
// src/services/SessionMemory/sessionMemory.ts
export async function extractSessionMemory(context: REPLHookContext): Promise<void> {
  // Only run on main REPL thread
  if (querySource !== 'repl_main_thread') return
  // Execute via forked subagent; does not block main conversation
  await runForkedAgent({ querySource: 'session_memory', ... })
}
```

5.3 Trigger Thresholds

```
// Initialization threshold (how far into the conversation before the first extraction)
minimumMessageTokensToInit: 1500  // token count

// Update threshold (after each extraction, minimum token growth before next extraction)
minimumTokensBetweenUpdate: 3000

// Tool call threshold (how many tool calls occur between extractions)
toolCallsBetweenUpdates: 50
```

5.4 Isolation Mechanism

Session Memory uses `runForkedAgent` to create an isolated context, preventing child processes from polluting parent process state:

```
// Isolate readFileState
overrides: { readFileState: setupContext.readFileState }
// Restrict tools: only allow Edit on specific memory files
canUseTool: createMemoryFileCanUseTool(memoryPath)
```

5.5 Integration with Compaction

```
// src/services/compact/sessionMemoryCompact.ts
// When context compression occurs, session memory content is also included in the
summarization process
```

---

6. Layer 5: ExtractMemories Service

> **Note**
>
> **See "Memory System" documentation for details**

6.1 Overview

The ExtractMemories service automatically extracts worth-persisting information from conversation history after each dialogue turn ends (stopHooks), writing it to the Auto Memory directory. Core file: `src/services/extractMemories/extractMemories.ts`.

6.2 Trigger Mechanism

```
// Called via handleStopHooks in stopHooks.ts
export async function executeExtractMemories(
  context: REPLHookContext,
  appendSystemMessage?: AppendSystemMessageFn,
): Promise<void>
```

6.3 Mutual Exclusion

Mutual exclusion strategy between ExtractMemories and the main Agent:

```
// If the main Agent just wrote a memory file itself, skip this extraction
if (hasMemoryWritesSince(messages, lastMemoryMessageUuid)) {
  return  // Skip to avoid duplication
}
```

6.4 Tool Permission Control

```
// src/services/extractMemories/extractMemories.ts
export function createAutoMemCanUseTool(memoryDir: string): CanUseToolFn {
  // Allow: Read / Grep / Glob (unrestricted)
  // Allow: Read-only Bash commands (ls, find, grep, cat, stat...)
  // Allow: Edit / Write only for files inside the auto-memory directory
  // Deny: All other tools
}
```

6.5 Extraction Throttling

```
// Controlled via GrowthBook feature flag tengu_bramble_lintel
// Default: extract once every 1 turn
turnsSinceLastExtraction < getFeatureValue_CACHED_MAY_BE_STALE('tengu_bramble_lintel', 1)
```

6.6 Trailing Run

If new messages arrive during extraction, the latest context is stashed, and a trailing extraction runs immediately after extraction completes:

```
if (inProgress) {
  pendingContext = { context, appendSystemMessage }
  // Execute trailing run after current run ends
}
```

---

7. Layer 6: Hooks System (Lifecycle Injection)

> **Note**
>
> **See "Hook System Implementation" documentation for details**

7.1 Overview

Hooks are shell commands or function callbacks configured by users in `.claude/settings.json`, triggered at specific events in Claude Code's lifecycle, with their results (JSON output) injected back into the conversation context. Core file: `src/utils/hooks.ts`.

7.2 Hook Types (by Trigger Timing)

```
type HookEvent =
  | 'PreToolUse'           // Before tool call
  | 'PostToolUse'          // After tool call
  | 'PostToolUseFailure'   // After tool call failure
  | 'PreCompact'           // Before compaction
```

```
  | 'PostCompact'          // After compaction
  | 'Stop'                 // When conversation stops
  | 'StopFailure'          // When conversation abnormally stops
  | 'PreExecute'           // Before user submission
  | 'UserPromptSubmit'     // When user submits
  | 'PermissionsChange'    // When permissions change
  | 'SessionStart'         // When session starts
  | 'SessionEnd'           // When session ends
  | 'SubagentStart'        // When subagent starts
  | 'SubagentStop'         // When subagent stops
  | 'TaskCreated'          // When task is created
  | 'TaskCompleted'        // When task is completed
  | 'ConfigChange'         // When config changes
  | 'InstructionsLoaded'   // When instruction file is loaded ← Important!
  | 'Elicitation'          // Before user selection
  | 'ElicitationResult'    // After user selection
  | 'ToolResult'           // When tool result is processed
```

7.3 InstructionsLoaded Hook (Conditional Injection)

This is the key Hook for implementing "inject exclusive memory upon entering a certain path":

```
// Defined in src/utils/hooks.ts
interface InstructionsLoadedHookInput {
  path: string           // Path of the loaded file
  type: MemoryType       // File type: 'User' | 'Project' | 'Local' | 'Managed'
  load_reason: string    // Load reason: 'session_start' | 'compact' | 'include'
  globs?: string[]       // Path patterns from frontmatter
  parentFilePath?: string // Parent file path when loaded via @include
}

// Triggered in claudemd.ts
void executeInstructionsLoadedHooks(file.path, file.type, loadReason, {
  globs: file.globs,
  parentFilePath: file.parent,
})
```

7.4 Hook Output Injection

Hook-returned JSON output is injected via:

```
// systemMessage / additionalContext fields in HookResult
interface HookResult {
  systemMessage?: string        // Injected as system message
  additionalContext?: string   // Appended to context
  blockingError?: string       // Block operation
  initialUserMessage?: string  // Replace user message
  updatedInput?: object        // Modify tool input
  // ...
}
```

7.5 asyncRewake Mechanism (Background Hook Wake-up)

Async hooks can run in the background, injecting results into the conversation via `queued_command` attachment:

```
// src/utils/hooks.ts
void shellCommand.result.then(async result => {
  if (result.code === 2) {
    // exit code 2 = blocking error, injected as task-notification
    enqueuePendingNotification({
      value: wrapInSystemReminder(`Stop hook blocking error: ${stderr || stdout}`),
      mode: 'task-notification',
    })
  }
})
```

7.6 Working Directory Trust Check

```
// src/utils/hooks.ts
export function shouldSkipHookDueToTrust(): boolean {
  // Non-interactive mode (SDK) skips trust check
  const isInteractive = !getIsNonInteractiveSession()
  if (!isInteractive) return false
  // Interactive mode: all hooks require accepting trust dialog before execution
  const hasTrust = checkHasTrustDialogAccepted()
  return !hasTrust
}
```

---

8. Layer 7: System Prompt Segmentation and Dynamic Boundaries

8.1 SYSTEM_PROMPT_DYNAMIC_BOUNDARY

`src/constants/system.ts` defines a key boundary marker:

```
// src/constants/prompts.ts
export const SYSTEM_PROMPT_DYNAMIC_BOUNDARY =
  '__SYSTEM_PROMPT_DYNAMIC_BOUNDARY__'
```

Purpose: Marks the dividing line between static content (cacheable across organizations) and dynamic content (user/session-specific) in the System Prompt. Content before the boundary uses `scope: 'global'` caching; content after cannot be cached.

8.2 Cache Breaker Injection

`src/context.ts` contains a system prompt injection mechanism for cache break debugging:

```
// src/context.ts
let systemPromptInjection: string | null = null

export function setSystemPromptInjection(value: string | null): void {
  systemPromptInjection = value
  // Clear context cache
  getUserContext.cache.clear?.()
  getSystemContext.cache.clear?.()
}
```

```
// In getSystemContext:
if (feature('BREAK_CACHE_COMMAND')) {
  injection = getSystemPromptInjection()
  // Injected as cacheBreaker
  return { cacheBreaker: `[CACHE_BREAKER: ${injection}]` }
}
```

> **Note**
>
> cacheBreaker is a controllable cache break switch, **for internal debugging only** — by injecting a unique identifier in every request's system context, engineers can precisely control when to break the Prompt Cache, enabling measurement and analysis of cache hit behavior

8.3 Cache Breaker Detection

`src/services/api/promptCacheBreakDetection.ts` monitors whether the prompt cache is being used correctly.

---

9. Layer 8: Conditional Rules Injection

9.1 Nested Memory Loading Mechanism

CLAUDE.md rules support two loading strategies:

1. **Eager Loading**: All rules are loaded at once at session start
2. **Nested/Demand Loading**: Conditional rules are loaded only when accessing specific files/directories — the method that triggers memories via frontmatter `paths` based on conditions

```
// src/utils/claudemd.ts
// Phase 1: getManagedAndUserConditionalRules - load global conditional rules
export async function getManagedAndUserConditionalRules(
  targetPath: string,
  processedPaths: Set<string>,
): Promise<MemoryFileInfo[]>

// Phase 2: getMemoryFilesForNestedDirectory - load nested directory rules
export async function getMemoryFilesForNestedDirectory(
  dir: string,
  targetPath: string,
  processedPaths: Set<string>,
): Promise<MemoryFileInfo[]>

// Phase 3: getConditionalRulesForCwdLevelDirectory - load CWD-level rules
export async function getConditionalRulesForCwdLevelDirectory(...)
```

9.2 Glob Pattern Matching

```
// src/utils/claudemd.ts
// Uses the ignore library (respects .gitignore semantics)
import ignore from 'ignore'
```

```
return ignore().add(file.globs).ignores(relativePath)

// Supported frontmatter format:
// paths:
//   - "src/**/*.ts"
//   - "src/**/*.tsx"
```

9.3 Base Path Resolution

Path globs have different base directories depending on rule type:

```
const baseDir =
  type === 'Project'
    ? dirname(dirname(rulesDir)) // Parent of .claude/rules/ = project root
    : getOriginalCwd()           // Managed/User rules relative to original CWD
```

Claude Code 的规则文件（.md）有三个来源，对应三个 type:

type	规则文件目录	典型路径
Managed	~/.claude/rules/	全局配置目录，存放统一管理的规则
User	~/.claude/rules/	用户个人目录
Project	<项目根>/.claude/rules/	随项目代码库一起版本控制的规则

- **Project type**: `rulesDir = /path/to/project/.claude/rules`, going up two levels gives `/path/to/project` — the project root. Project rules are written by team members under CLAUDE.md or .claude/rules/. The rule author's mental reference when describing globs is "project root", so `globs: ['src/**/*.ts']` is a path relative to the project root.
- **Managed/User types**: These rules are placed under `~/.claude/rules/`. Users may run claude in any directory. When the rule author writes globs, the "current directory" they reference should be the directory where the user started claude (i.e., `getOriginalCwd()`), not the home directory. So `globs: ['project/src/**/*.ts']` is understood as "relative to where I ran claude".

---

10. Post-Sampling Hook

Note

"Sampling" is Anthropic API terminology for **the process of the model generating a token from a probability distribution**. A complete "sample" is the model generating a full response — i.e., an assistant message containing all content blocks (text, tool_use, etc.)

Post-Sampling Hook 的时机

```
发送请求 → API 返回流 → 模型完成采样（回复生成完毕）
                              ↓
                        executePostSamplingHooks() ← 在这里执行
                              ↓
                        执行工具（tool_use 触发 CCR）
                              ↓
                        返回结果给模型（继续下一轮）
```

Timing: The model has fully generated a response, but has not yet started executing tools
Purpose: The core value of this type of hook is performing post-processing based on content the model just generated, without blocking the tool execution chain (doing extra work based on model output as a side effect)

10.1 Core Mechanism

`src/utils/hooks/postSamplingHooks.ts` defines an internal API that allows executing side effects after model sampling completes:

```
// Post-sampling hook registry (internal API, not exposed in settings.json)
const postSamplingHooks: PostSamplingHook[] = []

export function registerPostSamplingHook(hook: PostSamplingHook): void {
  postSamplingHooks.push(hook)
}

export async function executePostSamplingHooks(
  messages: Message[],
  systemPrompt: SystemPrompt,
  userContext: { [k: string]: string },
  systemContext: { [k: string]: string },
  toolUseContext: ToolUseContext,
  querySource?: QuerySource,
): Promise<void>
```

10.2 Usage Scenarios

Currently primarily used for automatic Session Memory extraction:

```
// src/services/SessionMemory/sessionMemory.ts
const extractSessionMemory = sequential(async function (context: REPLHookContext) {
  // Run only on main thread
  if (context.querySource !== 'repl_main_thread') return
  // Check GrowthBook gate (non-blocking, cached value)
  if (!isSessionMemoryGateEnabled()) return
  // Check token and tool call thresholds
  if (!shouldExtractMemory(messages)) return
  // Fork subagent to execute extraction
  await runForkedAgent({ querySource: 'session_memory', ... })
})

export function initSessionMemory(): void {
  registerPostSamplingHook(extractSessionMemory)
}
```

Currently registered items:

Hook	作用
extractSessionMemory	从模型输出中提取值得记忆的内容，写入 auto memory
updateMagicDocs	如果模型改动了 MagicDocs 对应的文件，更新文档缓存
createSkillImprovementHook	分析 skill 执行结果，改进 skill 质量
Prompt 推测 (speculation)	基于模型输出预判下一步可能用到的提示
/auto-dream	生成梦境记忆

Distinction from other hooks:

- Pre-Sampling Hooks (before-tool-call, etc.): run before/after tool execution; can modify request content or intercept tools
- Post-Sampling Hook: runs after model response is generated but before tool execution; used for side-effect analysis
- Stop Hook: runs when model stops mid-generation; can inject extra context to let it continue

---

11. appendSystemPrompt and overrideSystemPrompt

11.1 Two Layers of Extra Injection

`buildEffectiveSystemPrompt()` in `src/utils/systemPrompt.ts` accepts two extra prompt parameters:

```
export function buildEffectiveSystemPrompt({
  overrideSystemPrompt,  // Priority 0: completely replace all other prompts
  customSystemPrompt,     // Priority 3: custom prompt (via --system-prompt)
  appendSystemPrompt,     // Always appended at the end
  defaultSystemPrompt,
  // ...
}): SystemPrompt
```

11.2 Priority System

```
Priority 0: overrideSystemPrompt → complete replacement, return
asSystemPrompt([overrideSystemPrompt])

Priority 1: Coordinator mode → use coordinator prompt

Priority 2: Agent mode (mainThreadAgentDefinition)
  - PROACTIVE/KAIROS mode: agent prompt appended to default
  - Other modes: agent prompt replaces default

Priority 3: customSystemPrompt (--system-prompt CLI flag)

Priority 4: defaultSystemPrompt

Priority 5: appendSystemPrompt (always appended at the end, except with
overrideSystemPrompt)
```

Note

KAIROS/PROACTIVE Mode:

- KAIROS/PROACTIVE transforms Claude Code from a "tool that responds when called" into a "continuously running, proactively reporting assistant", achieved via the SendUserMessage tool and periodic tick mechanism, enabling user-driven continuous workflows
- **Continuous Running Mechanism**
  - When KAIROS is active and the message queue is idle, print.ts:2477 automatically injects a proactive tick, allowing the model to "wake up" and continue working without user input — such as checking background task status or pushing progress updates
- **Changes to Memory Mode**

- Under KAIROS, buildAssistantDailyLogPrompt replaces the standard memory prompt: instead of maintaining a MEMORY.md index, the model appends writes to daily log files (YYYY/MM/YYYY-MM-DD.md), which are consolidated into MEMORY.md by a nightly process. This suits long-running sessions and avoids frequent index file modifications

11.3 Cowork Extra Guidelines Injection

```
// Injected via environment variable
const coworkExtraGuidelines = process.env.CLAUDE_COWORK_MEMORY_EXTRA_GUIDELINES
// Injected into Auto Memory's buildMemoryLines
```

Note

- Cowork is a desktop/embedded mode for Claude Code: **running Claude Code embedded in a desktop application, rather than via terminal CLI**. Users can connect via the Cowork desktop client or Cowork SDK
- **Relationship**: Cowork is the embedder/consumer of Claude Code; Claude Code is the underlying engine layer

Key Differences Between Cowork and Normal CLI

- Message routing is controlled by the desktop application (Cowork bridges WebSocket, not REPL)
- Memory paths are externally controlled (injected via environment variables, not auto-inferred by git root)
- Plugin directory uses cowork_plugins instead of plugins
- Settings file uses cowork_settings.json instead of settings.json

Purpose of CLAUDE_COWORK_MEMORY_EXTRA_GUIDELINES
As an embedded mode, Cowork operators (e.g., enterprise IT) need to inject extra behavioral guidance for the model. This guidance is environment-level and not hardcoded. In other words, CLAUDE_COWORK_MEMORY_EXTRA_GUIDELINES allows Cowork operators to **inject extra behavioral specifications into the model's memory system at the environment variable level, without modifying Claude Code's own code**.

For example: an enterprise Cowork may require the model to "all code changes must comply with internal code standards" or "external communications must go through specific approval processes". These are injected via this environment variable:

```
const coworkExtraGuidelines = process.env.CLAUDE_COWORK_MEMORY_EXTRA_GUIDELINES
// Injected into the extraGuidelines parameter of the memory prompt
```

It is injected in two places:

- Auto Memory (loadMemoryPrompt): guidance for the memory system in normal sessions
- Agent Memory (agentMemory.ts): guidance for subagent memory systems

In buildMemoryLines, extraGuidelines are appended as a separate paragraph at the end of the memory instructions, so the model directly sees these extra rules when reading.

Purpose of CLAUDE_COWORK_MEMORY_PATH_OVERRIDE
In normal CLI mode, the memory directory is inferred by git root hash, with one directory per project. But in

Cowork, memory paths are controlled by external mounts (possibly user or team-scoped), so this environment variable directly overrides the path calculation logic, allowing Cowork operators to precisely control where memories are stored.

---

12. Coordinator and Agent Prompt Priority

12.1 Coordinator Mode

> **Note**
>
> **See "Agent Algorithm Flow" documentation for details**

When `feature('COORDINATOR_MODE') && process.env.CLAUDE_CODE_COORDINATOR_MODE` is set, the Coordinator Prompt is used:

```
// src/utils/systemPrompt.ts
if (feature('COORDINATOR_MODE') && isEnvTruthy(process.env.CLAUDE_CODE_COORDINATOR_MODE)) {
  return asSystemPrompt([getCoordinatorSystemPrompt(), ...appendSystemPrompt])
}
```

12.2 Agent Definition Injection

Built-in Agents (loaded by `loadAgentsDir.ts`) can provide per-agent-type system prompts:

```
// src/tools/AgentTool/loadAgentsDir.ts
const agentSystemPrompt = mainThreadAgentDefinition
  ? mainThreadAgentDefinition.getSystemPrompt()
  : undefined
```

---

13. Compact (Context Window Compression)

> **Note**
>
> **See "Context Management" documentation for details**

13.1 Compression Trigger Timing

```
// src/services/compact/autoCompact.ts
// Automatically triggered when context approaches token limit
// After triggering, old messages are replaced with summaries via summarizeMessages
```

13.2 Memory Handling During Compression

```
// src/services/compact/sessionMemoryCompact.ts
// During compression, Session Memory also participates in the summarization process
// The extractMemories service reloads memories after compression
(resetGetMemoryFilesCache('compact'))
```

13.3 Compact Boundary Marker

```
// src/utils/messages.ts
export function createCompactBoundaryMessage(
  partialCompactDirection?: PartialCompactDirection
): SystemCompactBoundaryMessage
// Inserts a boundary marker into the message sequence after compression, for subsequent
tracking
```

---

14. Architecture Summary and Key Design Patterns

> **Note**
>
> Beyond the above, some exceptional situations also use dynamic context injection. For example, when the model output reaches the `max_output_tokens` limit, the following prompt may be injected:
>
> ```
> content: `Output token limit hit. Resume directly — no apology, no recap. ` +
>  `Pick up mid-thought if that is where the cut happened. ` +
>  `Break remaining work into smaller pieces.`,
> ```

14.1 Layered Responsibility Overview

Layer	Mechanism	Trigger Timing	Persistence
System Reminder	`<system-reminder>` XML tag	Always (via message attachments)	No
CLAUDE.md	File loading	Session start + on-demand	No (files themselves persist)
Auto Memory	Guidance + memory files	Session start + extraction	Yes (disk files)
Session Memory	Forked subagent	Post-Sampling	Yes (session-level disk files)
ExtractMemories	Forked subagent	StopHooks	Yes (Auto Memory)
Hooks	Shell/callback	Lifecycle events	Varies by hook
Coordinator/Agent	Prompt replacement/append	CLI flag / feature	No
Compact	Summary replacement	Token threshold	No

14.2 Key Design Patterns

Pattern 1: Forked Agent (Isolated Execution)

All background operations that may produce side effects (Session Memory, ExtractMemories) are executed via `runForkedAgent()` in an isolated context, without polluting main conversation state:

```
// src/utils/forkedAgent.ts
await runForkedAgent({
  promptMessages,
  cacheSafeParams,        // Inherit parent's prompt cache
  canUseTool,              // Restrict tool permissions
```

```
  overrides: { readFileState: isolatedContext.readFileState }
})
```

Pattern 2: GrowthBook Feature Gate

Nearly all new mechanisms have GrowthBook feature flags for gradual rollout:

```
if (!getFeatureValue_CACHED_MAY_BE_STALE('tengu_passport_quail', false)) return
if (!getFeatureValue_CACHED_MAY_BE_STALE('tengu_session_memory', false)) return
```

Note

- Definition: GrowthBook is an A/B testing / feature flag platform that Claude Code uses for gradual feature rollout and remote control of code behavior, without redeployment
- Two usage modes
  - Build-time constant folding (`feature('FLAG_NAME')`): these flags are used for major product-line switching (KAIROS, BRIDGE_MODE, PROACTIVE, etc.), deciding at build time whether to include certain code
  - Runtime remote evaluation (`getFeatureValue_CACHED_MAY_BE_STALE`): these flags' values are fetched from the GrowthBook server and can be changed at any time, used for: 1) gradual rollout of a feature per user/organization; 2) real-time kill-switches for live features; 3) A/B testing different parameter values (e.g., thresholds, timeouts)
- **User attribute targeting**: GrowthBook determines each flag's return value based on user attributes (user type, subscription tier, org ID, etc.), enabling per-user-group control

Pattern 3: Memoize Caching

Large amounts of context data are cached via `lodash-es/memoize` to avoid recomputation:

```
export const getSystemContext = memoize(async () => { ... })
export const getUserContext = memoize(async () => { ... })
export const getMemoryFiles = memoize(async () => { ... })
```

Pattern 4: Cache Invalidation Propagation

When values that affect context change, all related caches are cleared in cascade:

```
export function setSystemPromptInjection(value: string | null): void {
  systemPromptInjection = value
  getUserContext.cache.clear?.()
  getSystemContext.cache.clear?.()
}
```

Pattern 5: System Prompt Segmented Caching

`systemPromptSection()` supports a `cacheBreak: boolean` flag, allowing certain segments to be recomputed every turn without breaking the entire prompt cache:

```
// Stable segment: cacheBreak: false (can be cached)
export function systemPromptSection(name: string, compute: ComputeFn): SystemPromptSection
```

```
// Dynamic segment: cacheBreak: true (recomputed every turn; will break prompt cache)
export function DANGEROUS_uncachedSystemPromptSection(
  name: string,
  compute: ComputeFn,
  _reason: string,  // Reason must be provided
): SystemPromptSection
```

Pattern 6: Timing-Aware Extract Mem

ExtractMemories works only when the main Agent does **not** produce memory writes (mutual exclusion), avoiding race conditions when writing to the same file.

Pattern 7: Memory Type Classification Constraints

Enforced classification via frontmatter `type` field, with detailed save/access/trust guidelines, preventing the memory system from being abused as general-purpose storage.

Appendix: Key File Index

File	Responsibility
`src/utils/messages.ts`	System Reminder wrapping, message normalization, compact boundary
`src/context.ts`	getSystemContext / getUserContext (git status, CLAUDE.md, date)
`src/utils/claudemd.ts`	Complete CLAUDE.md discovery, loading, parsing, caching pipeline
`src/memdir/memdir.ts`	Auto Memory prompt building, entrypoint truncation
`src/memdir/memoryTypes.ts`	Memory type classification, save/access/trust guidance text
`src/services/SessionMemory/sessionMemory.ts`	Session memory auto-extraction (Post-Sampling)
`src/services/extractMemories/extractMemories.ts`	Conversation history auto-extraction to persistent memory (StopHooks)
`src/utils/hooks.ts`	Complete Hook system (16+ event types)
`src/utils/hooks/postSamplingHooks.ts`	Post-sampling Hook mechanism
`src/constants/prompts.ts`	System Prompt section building
`src/constants/systemPromptSections.ts`	Segmented cache management
`src/utils/systemPrompt.ts`	Effective System Prompt priority building
`src/utils/queryContext.ts`	API cache key prefix building
`src/services/compact/compact.ts`	Context compression
`src/services/compact/autoCompact.ts`	Automatic compression triggering
`src/utils/dynamicConfig.ts` / `src/hooks/useDynamicConfig.ts`	Dynamic configuration (GrowthBook)